Exercises on Inequalities for Calculus

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Level 1 — Algebraic Inequalities

1. Prove that for every real number x,
$$x^2 + 1 \geq 2x.$$
2. Show that
$$(a-b)^2 \geq 0$$
implies
$$\frac{a+b}{2} \geq \sqrt{ab}, \qquad a, b > 0.$$
3. Prove that
$$|a+b| \leq |a| + |b|.$$
4. Prove that
$$\big||a| - |b|\big| \leq |a-b|.$$
5. If $0 < x < 1$, prove that
$$x^2 < x < \sqrt{x}.$$
6. If $x > 0$, prove that
$$x + \frac{1}{x} \geq 2.$$

Level 2 — Inequalities Involving Elementary Functions

7. Show that for every real number x,
$$|\sin x| \leq |x|.$$
8. Show that
$$1 - \cos x \leq \frac{x^2}{2}.$$
9. For every real number x, prove that
$$e^x \geq 1 + x.$$
10. Prove that for $x > -1$,
$$\ln(1+x) \leq x.$$
11. Show that for $x > 0$,
$$\ln x \leq x - 1.$$
12. Prove that for every real number x,
$$\cosh x \geq 1 + \frac{x^2}{2}.$$

Level 3 — Bounding Sequences

13. Prove that

$$\frac{1}{n+1} < \ln\left(1+\frac{1}{n}\right) < \frac{1}{n}.$$

14. Show that

$$\left(1+\frac{1}{n}\right)^n < e.$$

15. Prove that

$$\left(1+\frac{1}{n}\right)^n < \left(1+\frac{1}{n+1}\right)^{n+1}.$$

16. Show that

$$\sum_{k=1}^{n} \frac{1}{k} > \ln(n+1).$$

17. Prove that

$$\sum_{k=1}^{n} \frac{1}{k} < 1+\ln n.$$

Level 4 — Integral Inequalities

18. If $f(x) \geq 0$ on $[a, b]$, prove that

$$\int_a^b f(x)\, dx \geq 0.$$

19. If

$$f(x) \leq g(x)$$

on $[a, b]$, prove that

$$\int_a^b f(x)\, dx \leq \int_a^b g(x)\, dx.$$

20. Prove that

$$\sin x < x < \tan x, \qquad 0 < x < \frac{\pi}{2}.$$

21. Show that

$$\frac{x}{1+x} \leq \ln(1+x) \leq x, \qquad x > 0.$$

22. Let $f : [a, b] \to \mathbb{R}$ satisfy $f(x) > 0$ for all $x \in [a, b]$. Prove that

$$\frac{b-a}{\max f} \leq \int_a^b \frac{1}{f(x)}\, dx.$$

23. Let f be integrable on $[a, b]$. Prove the triangle inequality for integrals:

$$\left|\int_a^b f(x)\, dx\right| \leq \int_a^b |f(x)|\, dx.$$

24. Let f, g be integrable on $[a, b]$. Prove the Cauchy–Schwarz inequality for integrals:

$$\left(\int_a^b f(x)g(x)\, dx\right)^2 \leq \left(\int_a^b f(x)^2\, dx\right)\left(\int_a^b g(x)^2\, dx\right).$$

Level 5 — Inequalities Used in Convergence Proofs

25. Determine for which values of p,
$$\frac{1}{n^p} \le \frac{1}{n}.$$

26. Show that
$$\frac{n}{n^2+1} < \frac{1}{n}.$$

27. Find a constant C such that, near the origin,
$$|\sin x - x| \le C|x|^3.$$

28. Prove that
$$\left|\frac{\sin x}{x}\right| \le 1, \qquad x \neq 0.$$

29. Show that
$$\frac{1}{n+1} < \ln(n+1) - \ln n < \frac{1}{n}.$$

30. Use inequalities to prove that
$$\lim_{n\to\infty} \frac{\ln n}{n} = 0.$$

Level 6 — Classical Inequalities

31. Prove the Arithmetic Mean–Geometric Mean (AM–GM) inequality: for nonnegative real numbers $a_1, \dots, a_n$,
$$\frac{a_1 + a_2 + \cdots + a_n}{n} \ge \sqrt[n]{a_1 a_2 \cdots a_n}.$$

32. Prove the Cauchy–Schwarz inequality: for real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,
$$\left(\sum_{i=1}^n a_i b_i\right)^2 \le \left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right).$$

33. Prove Hölder's inequality: for $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$,
$$\sum_{i=1}^n |a_i b_i| \le \left(\sum_{i=1}^n |a_i|^p\right)^{1/p} \left(\sum_{i=1}^n |b_i|^q\right)^{1/q}.$$

34. Prove Minkowski's inequality: for $p \ge 1$,
$$\left(\sum_{i=1}^n |a_i + b_i|^p\right)^{1/p} \le \left(\sum_{i=1}^n |a_i|^p\right)^{1/p} + \left(\sum_{i=1}^n |b_i|^p\right)^{1/p}.$$

35. Prove Jensen's inequality: if f is convex, then for $x_1, \dots, x_n$,
$$f\left(\frac{1}{n}\sum_{i=1}^n x_i\right) \le \frac{1}{n}\sum_{i=1}^n f(x_i).$$

36. Prove Bernoulli's inequality:

$$(1+x)^r \geq 1 + rx,$$

for the appropriate values of r and x.

37. Prove Young's inequality:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}, \qquad \frac{1}{p} + \frac{1}{q} = 1.$$

38. Prove the rearrangement inequality: if $a_1 \leq a_2 \leq \cdots \leq a_n$ and $b_1 \leq b_2 \leq \cdots \leq b_n$, then for any permutation σ of $\{1, \ldots, n\}$,

$$\sum_{i=1}^{n} a_i b_{n+1-i} \leq \sum_{i=1}^{n} a_i b_{\sigma(i)} \leq \sum_{i=1}^{n} a_i b_i.$$

39. Prove Chebyshev's sum inequality: if $a_1 \leq a_2 \leq \cdots \leq a_n$ and $b_1 \leq b_2 \leq \cdots \leq b_n$, then

$$\frac{1}{n} \sum_{i=1}^{n} a_i b_i \geq \left(\frac{1}{n} \sum_{i=1}^{n} a_i \right) \left(\frac{1}{n} \sum_{i=1}^{n} b_i \right).$$

40. Prove the power mean inequality: for positive reals $a_1, \ldots, a_n$ and nonzero exponents $r < s$, the power means

$$M_r = \left(\frac{1}{n} \sum_{i=1}^{n} a_i^r \right)^{1/r}$$

satisfy

$$M_r \leq M_s.$$

41. Prove Nesbitt's inequality: for positive reals a, b, c,

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Level 7 — Inequalities from Taylor's Theorem

42. Let f be $(n+1)$-times differentiable on an interval containing a and x, with $|f^{(n+1)}(t)| \leq M$ for all t between a and x. Prove Taylor's inequality:

$$\left| f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \right| \leq \frac{M}{(n+1)!}|x-a|^{n+1}.$$

43. Use Taylor's inequality to show that for $0 \leq x \leq 1$,

$$\left| e^x - \left(1 + x + \frac{x^2}{2} \right) \right| \leq \frac{e}{6} x^3.$$

44. Use Taylor's inequality to show that for every real number x,

$$\left| \sin x - \left(x - \frac{x^3}{6} \right) \right| \leq \frac{x^4}{24}.$$